

PROJECT PHOENIX

Product Analytics for Zomato Bangalore Dataset

SQL • MySQL • Power BI — Customer Funnel, Retention & Restaurant-Marketplace Analysis

Personal Project | Prepared by Sheetal



Customers	Sessions	Orders	Revenue	AOV	Conversion
10,000	~30,000	5,903	~₹5,000,000	₹858.61	19.68%

Table 0. Headline KPIs from the finished Power BI dashboard (dashboard-level figures are rounded; underlying measures were used for exact calculations).

1. Project Objective

Project Phoenix set out to analyze customer acquisition, conversion, ordering behaviour and restaurant-marketplace characteristics for a simulated food-delivery platform, in order to surface actionable, evidence-backed product insights. The guiding business question was:

“What factors are associated with customers failing to return to a food-delivery platform, and where can the product team intervene?”

This later sharpened into a more specific, testable problem statement once the funnel and retention analysis was complete:

“How can a food-delivery platform increase second-order conversion among first-time customers to improve customer retention and revenue?”

The project was scoped around five pillars: SQL data work, funnel and cart-abandonment analysis, retention analysis, root-cause analysis (RCA), and product recommendations — deliberately excluding machine learning, churn prediction, and other scope-creep additions that would not add proportional interview value.

2. Data & Methodology

2.1 Data architecture: real + simulated

The project combines a real, publicly sourced Zomato Bangalore restaurant dataset with a simulated behavioural layer for customers, sessions, carts and orders. The transactional/behavioural layer is explicitly simulated for product-analytics practice and is documented as such — it is never presented as proprietary Zomato transaction data.

- **Real data:** restaurant listings (name, location, cuisine, cost, rating, votes, online-order flag, restaurant type).
- **Simulated data:** customers, sessions, cart events, orders, order items, payments, delivery, reviews and coupons, generated with realistic drop-off rules rather than random numbers.

This architecture was a deliberate choice: transaction-level customer data isn't publicly available for a marketplace like Zomato, so the honest and technically defensible approach was real restaurant data + an explicitly simulated behavioural layer + SQL-based product analysis, with every simulated finding framed as an observed pattern in the simulated dataset rather than a claim about real Zomato users.

2.2 Cleaning the restaurant dataset

The raw restaurant dataset started at 51,717 rows. Repeated UTF-8 encoding corruption was identified and repaired in Python, after which the dataset was reduced to 8 analytically useful fields: restaurant name, online-order availability, rating, votes, location, restaurant type, cuisines, and approximate cost for two people.

Step	Result
Raw rows	51,717
Encoding issue	Corrupted UTF-8 characters identified and repaired
Columns retained	8 (name, online_order, rating, votes, location, rest_type, cuisines, approx_cost(for two))
Clean rows loaded to DB	41,263
Missing values after cleaning	0

Table 1. Restaurant-data cleaning pipeline. Duplicate-looking rows in the 8-column view were deliberately retained, since the full 17-column source had zero exact duplicates — the apparent duplicates only emerged after dropping columns, so removing them risked discarding real observations.

2.3 Simulated behavioural layer

A synthetic customer/order ecosystem was built in Python (Pandas/NumPy) on top of the cleaned restaurant dimension, using realistic stage-by-stage drop-off assumptions for the funnel (Session → Search → Restaurant View → Cart → Checkout → Payment → Order) rather than uniform random generation. The core tables used for analysis were customers, sessions, orders and the cleaned restaurants table, loaded into a relational database (MySQL) for SQL-based analysis.

- **Customers:** 10,000
- **Sessions:** ~30,000, each carrying an acquisition channel
- **Orders:** 5,903, each linked to a customer, session, order value, delivery time and rating

Generated data was validated for internal consistency (e.g., checking that funnel and cancellation rates were plausible relative to each other) before being used for analysis.

2.4 SQL analysis approach

Roughly 25–35 targeted SQL analyses were written against the database, deliberately favouring meaningful business questions over volume. Techniques used included joins, CTEs, window functions, CASE-based bucketing, date functions and aggregation — covering:

- Basic KPIs: total orders, revenue, AOV, active customers
- Intermediate: acquisition-channel ranking, restaurant ranking, repeat-customer identification
- Advanced: cohort/retention analysis, funnel and cart-abandonment analysis, customer segmentation by order frequency

2.5 Analytical questions

- Which acquisition channels generate the most orders and revenue, and do they differ in retention?
- Where does the customer funnel lose the most users?
- How do orders and revenue change over time?
- What share of customers return for a repeat order, and how quickly?
- Does restaurant popularity (votes) relate to rating, and does price relate to online-order adoption?
- How common is online ordering across restaurants, locations, cuisines and restaurant types?

3. Data Model

The schema was designed around 4 tables — enough to genuinely support funnel, cart-abandonment and retention analysis, without becoming an over-engineered “25-table monster.” Delivery partners were deliberately excluded as a separate table, since partner-level analysis wasn't needed for the project's core questions.

Table	Source	Purpose
restaurants_clean	Real (public dataset)	Restaurant, cuisine, location and cost attributes
customers	Simulated	Customer profiles
sessions	Simulated	App activity / acquisition channel
orders	Simulated	Transactions

Table 2. Data model — 4 tables spanning the real restaurant dimension and the simulated behavioural layer.

Conceptually, the restaurant dimension (real data) sits alongside the simulated behavioural layer (restaurants_clean, customers, sessions, orders), all loaded into a single relational database and connected through customer_id, session_id, order_id and restaurant_id foreign keys, so that funnel, retention and marketplace questions can all be answered with joins against the same core tables.

4. Dashboard Explanation

Two Power BI dashboards were built, kept deliberately lean (an executive/product overview and a marketplace view) rather than a large number of overlapping pages.

4.1 Dashboard 1 — Customer & Order Performance

This dashboard brings together headline KPIs, acquisition-channel performance, the conversion funnel, monthly order/revenue trends and rating distribution. It is designed as an executive/product overview rather than a detailed operational dashboard.

- **Headline KPIs:** Count of orders (5.903K), Sum of order value (5M), Average order value (858.61), Count of customers (10K), Conversion rate (20%), Count of sessions (30K).
- **Orders by Acquisition Channel / Orders vs Revenue by Channel:** compares order volume and revenue contribution across channels.
- **Revenue by Acquisition Channel:** bar chart of revenue contribution per channel.
- **Customer Conversion Funnel:** funnel visual from session to order, used to spot the largest drop-off stage.
- **Revenue Over Time / Orders Over Time:** monthly trend lines showing that revenue and order volume fluctuate rather than growing steadily.
- **Orders by Rating:** distribution of orders across restaurant rating bands.
- **Conversion Rate by acquisition channel:** bar chart showing that conversion is broadly similar (~19–20%) across channels.

Values shown as 5M revenue and 20% conversion are dashboard-level rounded values; underlying DAX measures were used for exact calculations (AOV = ₹858.61, conversion = 19.68%).

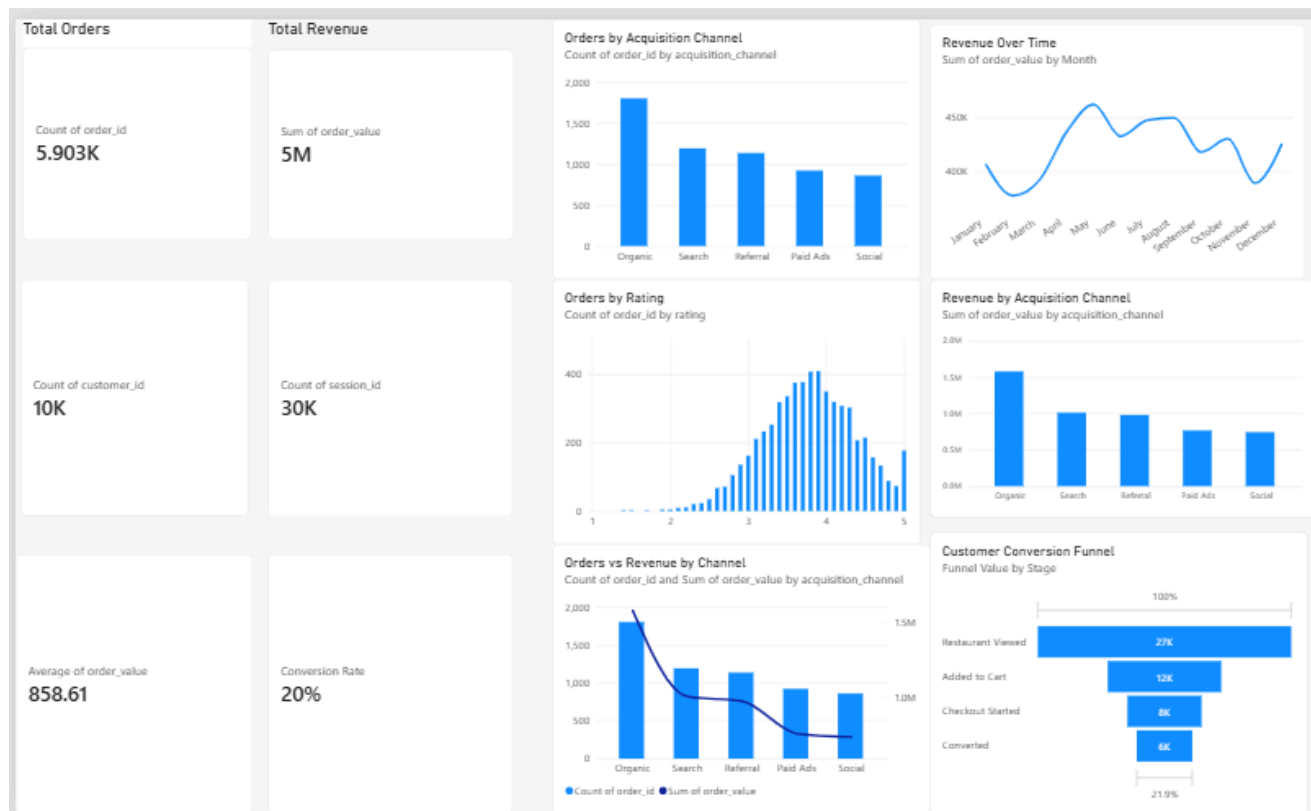


Fig : Static dashboard screenshots for the report

4.2 Dashboard 2 — Restaurant Marketplace Analysis

The marketplace dashboard uses restaurant-level data to examine distribution, popularity, rating, cost and online-order availability, intentionally limited to the most decision-relevant views.

- **Restaurant Distribution by Location:** bar chart of restaurant counts across the top locations (BTM, Koramangala 5th Block, HSR, Indiranagar, JP Nagar, and others).
- **Restaurant Cost vs Rating:** scatter plot of approximate cost for two against average rating.
- **Restaurant Rating vs Votes:** scatter plot of vote volume against average rating, by restaurant.
- **Restaurants Offering Online Ordering:** donut chart — 27,102 restaurants (65.68%) support online ordering vs. 14,161 (34.32%) that do not, out of 41,263 total.

4.3 Supporting SQL analysis: funnel and retention

Alongside the two Power BI dashboards, the SQL analysis layer additionally investigated retention and repeat-order behaviour; these findings are presented in the report rather than as a separate Power BI page.



Fig 2 : Static dashboard screenshots for the report

5. Restaurant Marketplace Analysis

5.1 Marketplace scale

41,263 restaurants across 92 locations, 2,367 cuisine tags and 87 restaurant types. Overall average rating is 3.70/5 and average cost for two is ₹603.91.

5.2 Supply concentration by location

Location	Restaurants	Marketplace Share
BTM	3,879	9.40%
Koramangala 5th Block	2,297	5.57%
HSR	1,993	4.83%
Indiranagar	1,800	4.36%
JP Nagar	1,710	4.14%

Table 3. Top-5 locations by restaurant count. BTM alone accounts for 9.4% of the marketplace, more than any other single location.

5.3 Online-order adoption: geography, restaurant type, cuisine and price

Overall marketplace adoption is 65.68%, but this masks large variation across every dimension examined:

- **By location:** 18.02% (Majestic) to 94.27% (Nagawara); HSR reaches 83.24% across 1,993 restaurants vs. MG Road's 45.02% across 793 restaurants.
- **By restaurant type:** 4.66% (Fine Dining) to 82.66% (Beverage Shop) among types with meaningful restaurant counts, delivery-oriented formats adopt online ordering far more than premium dine-in formats (Bar 16.09%, Lounge 12.07%).
- **By cuisine:** highest for Burger (82.22%), Seafood (82.60%), Beverages (79.07%), Mughlai (77.49%); lowest for Continental (52.88%) and Asian (54.58%).
- **By price segment:** adoption peaks at ₹600–999 (72.99%) and falls to 37.27% at ₹1,500+, even though the ₹1,500+ segment has the highest average rating (4.16).

Price Segment	Restaurants	Avg Rating	Online-order Adoption
Under ₹300	7,285 (17.66%)	3.57	60.60%
₹300–599	17,881 (43.33%)	3.59	71.89%
₹600–999	9,447 (22.89%)	3.76	72.99%
₹1,000–1,499	3,988 (9.66%)	4.00	48.77%
₹1,500+	2,662 (6.45%)	4.16	37.27%

Table 4. Price segment vs. restaurant count, rating and online-order adoption. ₹300–599 is the dominant segment (43.33% of the marketplace); adoption and price show a non-linear relationship.

5.4 Pricing geography

- **Lavelle Road** is the most expensive location (₹1,365 average cost for two, 4.14 rating); Race Course Road (₹1,321) and MG Road (₹1,245) are also premium markets.
- **Brigade Road** is a notable exception: 1,052 restaurants but only ₹699 average cost — a large marketplace does not necessarily mean premium pricing.

5.5 Popularity, rating and cuisine quality

Popularity (votes)	Restaurants	Avg Rating
Under 100	5,745	3.48
100–499	2,386	3.81

Popularity (votes)	Restaurants	Avg Rating
500–999	566	4.05
1,000–4,999	451	4.26
5,000+	31	4.56

Table 5. Average rating rises consistently with review volume, from 3.48 (<100 votes) to 4.56 (5,000+ votes).

The most-reviewed restaurants (e.g., Byg Brewski Brewing Company – 16,832 votes, 4.9 rating; Toit – 14,956 votes, 4.7; Truffles – 14,726 votes, 4.7 at ₹900) show that popularity is not simply a function of being expensive, both affordable and premium restaurants achieve strong marketplace popularity.

Cuisine also differs in price and quality: Asian restaurants average ₹1,288 at a 4.14 rating, versus Street Food at ₹302 and a 3.62 rating, different cuisine segments represent very different combinations of supply, price, quality and digital adoption.

All marketplace relationships above (price↔rating, votes↔rating, type↔adoption) are reported as observed associations in this dataset, not causal claims — restaurant type, location and price are likely confounded with each other.

6. Key Insights

6.1 Funnel: the biggest drop-off is restaurant view → cart

Funnel Stage	Users	Conversion from Previous Stage
Sessions	30,000	—
Restaurant Viewed	26,944	89.8%
Added to Cart	12,065	44.8%
Checkout Started	7,885	65.4%
Converted / Order	5,903	74.9%

Table 6. Customer conversion funnel. Overall conversion (sessions → orders) = 19.68%.

The largest bottleneck by far is Restaurant View → Add to Cart: only 44.8% of restaurant viewers add an item to cart, a loss of 14,879 sessions, the single biggest opportunity in the funnel.

Acquisition channel shows limited variation in session-level conversion (all channels convert at roughly 19–20%), meaning acquisition-channel optimization alone is unlikely to move the needle as much as fixing the view-to-cart bottleneck.

6.2 Retention is the core opportunity

- **72.44% of ordering customers (3,209 of 4,430) are one-time customers;** only 27.56% (1,221) place a repeat order.
- **Repeat customers are disproportionately valuable:** 27.56% of customers generate 45.12% of revenue (₹22.87L vs. ₹27.82L from one-time customers, 54.88%).

- **The 1 → 2 order transition is the single sharpest opportunity:** 1,006 customers placed exactly 2 orders and already generated ₹17.30L (34.13% of revenue), moving even a portion of the 3,209 one-time customers into this group is more impactful than chasing highly frequent (3+ order) customers, who contribute only 10.98% of revenue combined.
- **Time to second order:** repeat customers took an average of 55.97 days between their first and second order (range 0–172 days), suggesting a long re-engagement window.
- **Retention by acquisition channel:** Paid Ads (29.15%) and Referral (28.71%) show the highest repeat rates, Search the lowest (26.00%), a modest 3.15-point spread, not enough to say channel drives retention.

6.3 Obvious explanations were tested and ruled out

One-time and repeat customers were compared on first-order value, delivery time and rating, and the differences were negligible:

Metric	One-time	Repeat	Difference
Customers	3,209	1,221	—
Avg order value	₹866.85	₹852.97	−₹13.88
Avg delivery time	39.71 min	39.89 min	+0.18 min
Avg rating	3.78	3.81	+0.03

Table 7. One-time vs. repeat customer experience metrics — the differences are too small to explain the retention gap.

This is a deliberately disciplined finding: delivery speed, order value and rating do not meaningfully differ between one-time and repeat customers in this dataset, so the analysis avoided inventing a service-quality explanation for churn. The evidence instead points to customer re-engagement and second-order activation as the product lever, not basic service quality.

6.4 Restaurant marketplace is structurally heterogeneous

- Restaurant popularity does not guarantee a higher rating individually, but in aggregate, more-reviewed restaurants show meaningfully higher average ratings (3.48 → 4.56 across popularity buckets).
- Online ordering is widespread (65.68%) but highly uneven, varying from ~18% to ~94% by location and from ~5% to ~83% by restaurant type.
- Higher-price restaurants tend toward somewhat higher ratings, but online-order adoption peaks in the mid-price segment (₹600–999) rather than rising with price.

7. Product Recommendations

Recommendations are framed as product hypotheses to test, grounded in the analysis above — not as claims the dataset has already proven.

1. Drive first-to-second-order conversion

- Post-first-order re-engagement at Day 7 / Day 14 (or personalized timing) to prompt reordering.
- Targeted second-order incentives (e.g., ₹X off, free delivery, or a personalized offer) aimed selectively at first-time customers rather than blanket discounting.

- Personalized restaurant/cuisine recommendations based on first-order behaviour.

2. Reduce restaurant-view → cart friction

- Clearer restaurant information, stronger visibility of ratings/reviews, and upfront delivery-ETA and pricing transparency on the restaurant page.
- Personalized restaurant recommendations to reduce decision friction at the largest funnel drop-off.

3. Prioritize acquisition channels by incremental value, not orders alone

- Compare channel volume, conversion and repeat rate together — Paid Ads and Referral show the strongest repeat rates and may warrant relatively higher investment, though the spread across channels is modest.

4. Segment merchant digital-enablement strategy by marketplace segment

- Online-order adoption varies sharply by restaurant type, cuisine and price band, so a single onboarding strategy is unlikely to work — premium/fine-dining and bar/lounge formats likely need a different digital-enablement approach than quick-bites and beverage shops.

5. Use popularity and rating together for restaurant discovery

- Surface a mix of highly rated + highly reviewed restaurants and highly rated “hidden gems” with fewer reviews, rather than ranking on rating or votes alone.

8. Limitations

- The customer, session, cart and order layer is entirely simulated; it demonstrates analytical methodology but cannot establish real Zomato customer behaviour, and no finding from it should be read as a claim about actual Zomato users.
- The analysis is descriptive/associative, not causal, relationships such as price ↔ rating, votes ↔ rating, and channel ↔ retention are observed associations, not proven causes, and may be confounded by restaurant type, location or other unobserved factors.
- Order timing in the simulated data is date-level, not timestamp-level, so exact time-to-second-order figures (e.g., the 0-day minimum) should not be over-interpreted.
- Cuisine counts are tag-level, not restaurant-level (a restaurant can carry multiple cuisine tags), so cuisine totals do not sum to the restaurant count of 41,263.
- Dashboard-level figures (e.g., 5M revenue, 20% conversion) are rounded for display; underlying DAX measures were used for exact calculations.

9. Future Work

- Incorporate real event-level data if/when available, including delivery time, cancellations, coupon redemption, and true repeat-order and cohort-retention behaviour.
- Build a dedicated Retention & Customer Journey dashboard page in Power BI (orders-per-customer, revenue-by-customer-type, time-to-second-order, repeat-rate-by-channel) to complement the two existing dashboards.
- Test the product hypotheses above via A/B testing once a real interaction layer exists (e.g., second-order incentive experiments, restaurant-page redesign for the view-to-cart step).

- Explore predictive churn / propensity-to-reorder modelling as a natural next step once the descriptive retention story is validated against real data.
- Control for restaurant type and location when re-examining price↔adoption and price↔rating relationships, to separate genuine effects from confounding.

10. Tools Used

Category	Tools
Database	MySQL
Data cleaning & simulation	Python (Pandas, NumPy)
Analysis	SQL — joins, CTEs, window functions, CASE, aggregation, date functions
Dashboarding	Power BI (DAX, data modeling)
Version control	Git / GitHub

Table 8. Tooling used across the data pipeline, from raw data to dashboard.

11. Sources:

Data Source - <https://www.kaggle.com/datasets/absin7/zomato-bangalore-dataset?resource=download>

Prepared as a personal Product Analytics case study, combining SQL-based analysis with Power BI dashboarding.